CSEC 3 Cloud Computing

PROGRAM DOCUMENTATION

Name: Muit, Ivy Pauline
Panliboton, Ayelyn Janne F.
Sanglay, Rhea Lizza

Yr/Section: BSCS 3A

NyanScape Cloud Architecture and Cost Estimate

Architecture Summary

NyanScape is a cloud-based cat community blogging platform where users can create posts, upload images, interact with other users, and communicate through messaging features.

The system frontend was developed using React and deployed through Vercel. This allows users to access the platform through web browsers and mobile devices with a responsive interface.

The backend uses Flask API hosted on Azure App Service. The backend handles user requests such as:

- User authentication
- Post creation
- Comment management
- File uploads
- API communication

For authentication and user management, Supabase was integrated into the system. Supabase manages user login, registration, and JWT authentication tokens securely. Uploaded cat images and files are stored in Azure Blob Storage to separate media storage from the main application logic. Azure AI Vision is used for basic image analysis to detect cat-related content in uploaded images. Azure Application Insights was added for monitoring system performance, logs, and error tracking.

Overall System Flow

Users → React Frontend (Vercel) → Flask Backend (Azure App Service) → Azure Services and Supabase

Azure Resources Used

Resource	Purpose
Azure App Service	Hosts the Flask backend API
Azure Blob Storage	Stores uploaded cat images and files
Azure AI Vision	Performs image analysis
Azure Application Insights	Monitors performance and logs
Supabase	Handles authentication and user data
Vercel	Hosts the React frontend

Estimated Monthly Cost Breakdown

The following estimates were based on the Azure Pricing Calculator using low-scale student or prototype deployment settings.

Azure Resource	Estimated Monthly Cost
Azure App Service (Basic B1)	$0/month
Azure Blob Storage	$2.04/month
Azure AI Vision	$0/month
Azure Application Insights	$0/month

Estimated Total Cost: **Approximately $2.04 per month**

Azure Pricing Calculator Screenshot

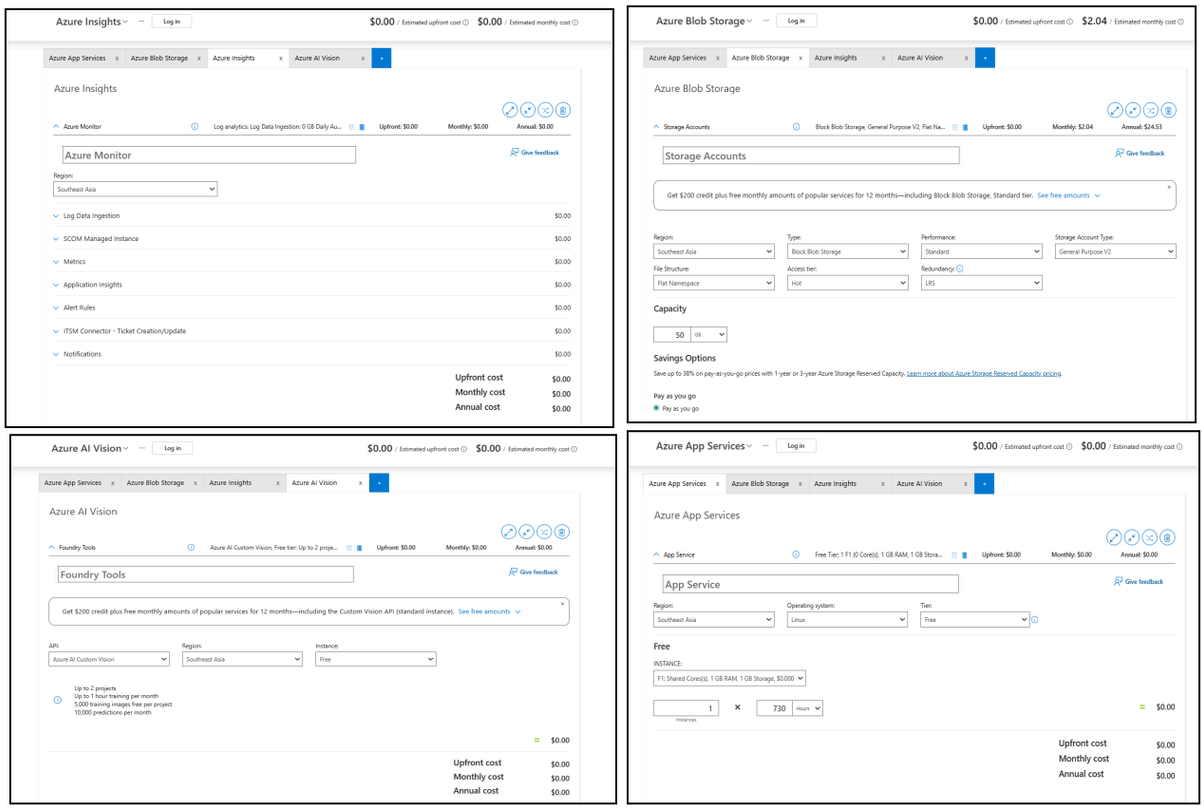

Figure 1. Azure Pricing Calculator Estimate for NyanScape

Cost Optimization Notes

Several strategies can be used to reduce the operational cost of the system.

- First, the project uses free-tier services from both Supabase and Vercel, which significantly lowers deployment expenses during development and testing.
- Second, Azure App Service can be scaled down during off-hours or periods of low traffic to reduce compute costs.

Additionally, Azure Blob Storage was used instead of storing images directly in the database because blob storage is more cost-efficient for media files.

In future deployments, reserved instances and auto-scaling features could also help reduce long-term cloud expenses while maintaining performance and reliability.

Conclusion

The NyanScape system demonstrates how modern cloud technologies can be combined to build a scalable and responsive social media platform. By integrating Azure services, Supabase authentication, and Vercel hosting, the system provides reliable performance while maintaining affordable monthly deployment costs suitable for student projects and small-scale applications.